

Waner (Arina) Zheng

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EDUCATION

Cornell University, College of Engineering, New York, United States
Master of Engineering in Financial Engineering, **GPA: 3.90**

Dec. 2025

Zhejiang University, Hangzhou, China

Bachelor of Economic Sciences in Finance, **GPA: 3.92**

Jun. 2024

Awards: Outstanding Graduate, Provincial Government Scholarship, Zhejiang University First Prize Scholarship

Coursework: Probability, Statistics, Stochastic Calculus, Optimization, Monte Carlo Simulation, Econometrics, Data Structures & Algorithms, Machine Learning, Reinforcement Learning, Algo Trading, Quant Risk Management, Derivatives, Fixed Income

Technical Skills: Python, SQL, C++, C#, R, MATLAB, Bash; Excel, Tableau, Stata, EViews; LaTeX

WORK EXPERIENCE

Data Scientist Intern, *AllianceBernstein LP*, New York, United States

Jun. to Aug. 2025

- **Factor Research:** Built a DeepSeek-LLM pipeline parsing news and 10-K/10-Qs on tech stocks to extract value, sentiment, risk, and growth factors with confidence scores; backtested and achieved 1.5% 1M alpha vs. S&P 500 in a long-short strategy.
- **Validation:** Developed an AI label consistency check framework, with only 2.2% inconsistency across ~17k predictions.
- **Trading System:** Implemented a multi-agent trading engine fusing LLM-based factors with price-volume indicators, applying volatility-targeted sizing, VaR limits, and stop-loss/take-profit rules; achieved 12.5% return with Sharpe 1.86 (2023–2024).

Quantitative Research Intern, *QTG Capital Management*, Shanghai, China

May to Jul. 2025

- **Factor Framework:** Built a scalable framework to generate hourly factors from OTM ETF option surfaces, transforming cross-sectional data into time-series signals and ranking them with rolling windows to create predictive timing strategies.
- **Performance:** Constructed separate long and short factor pools, retaining 200+ factors through tailored filters; long-only factors achieved 10% average return per trade, while short-only factors achieved Sharpe of 2.5 across 9 ETFs over 8 years.
- **ML Pipeline:** Developed ML pipelines using PCA-based feature selection, iterative labeling, and grid search, improving 1-hour directional prediction accuracy from 0.35 to 0.60; validated by cumulative return stratification and HMM regime analysis.

Quantitative Research Intern, *Soochow Securities Co., Ltd.*, Shanghai, China

Mar. to Aug. 2024

- **Alpha Research:** Developed 30+ novel price-volume factors using *Cutlets* segmentation, cross-sectional standardization, and correlation analysis, achieving ICIR values up to ~2.4.
- **Strategy Backtesting:** Built a Python decile long-short hedging framework delivering 20%+ annualized return; improved risk-adjusted performance with Random Forest, raising Sharpe ratio from 1.0 to 1.2.

Derivatives Trading Analyst Intern, *Zhejiang Zheqi Industrial Co., Ltd.*, Hangzhou, China

Jul. to Oct. 2023

- **Volatility Trading:** Developed and backtested a proprietary options trading strategy to exploit market maker behavior during options sweep events, identifying low-IV/high-vega contracts ahead of implied volatility repricing.
- **Data Engineering:** Utilized SQL to process and organize millisecond-level futures and options data in DBeaver databases.
- **Risk Monitoring:** Tracked and analyzed order cancellations, execution latency, and PnL exposure with Excel dashboards.

PROJECTS

Currency Market Forecasting and Analysis, *Barclays*, New York, United States

Aug. to Dec. 2025

- **Volume Forecasting:** Researched and implemented volume forecasting models with Hawkes processes and high-frequency FX trade data to predict transaction flows across major currency pairs (EUR/USD, JPY/USD, GBP/USD, MXN/USD).
- **Tool Development:** Developed an interactive Python tool integrating economic release calendars and model outputs, enabling Barclays' automated market-making desk to analyze deviations from forecasts and support real-time trading decisions.

Quantitative Risk Modeling Project, *Zhejiang University*, Hangzhou, China

Jan. to Feb. 2023

- **Risk Modeling:** Replicated Barra CNE5 model with CSI 500 equity data, estimating factor returns through cross-sectional regressions and constructing factor/specific risk components for portfolio risk decomposition.
- **Optimization & Validation:** Designed CVXPY-based minimum-variance portfolios under return targets, achieving 7–15% lower risk forecast errors and 12–20% greater stability out-of-sample, measured by realized vs. predicted volatility.

INTERESTS

Bel canto (8 years); Opera; Choir; Poker